



Santosh Barik

Date of birth: 31/03/2001 | **Nationality:** Indian | **Gender:** Male | **Phone:** (+91) 8917380324 (Mobile) | **Email address:** santoshbarik.phy@gmail.com | **LinkedIn:** www.linkedin.com/in/santosh-barik-706479209 | **Github:** <https://github.com/sbanty/qml-oam-timestamp> | **Whatsapp Messenger:** +91 9114057404 | **Address:** Plot No.- 488, Sahidnagar, Bhubaneswar, Odisha, 751007, Bhubaneswar, India (Home)

WORK EXPERIENCE

01/12/2022 - 14/11/2024 - BHUBANESWAR, INDIA

WEB DESIGNER AND DATA ANALYST DIGITAL DETA

Customized WordPress websites, handled coding and plugin integration, ensuring user-friendly and visually appealing interfaces. Analyzed data to provide insights, optimize performance, and support business decision-making.

15/05/2023 - 03/07/2023

SUMMER INTERNPHYSICAL RESEARCH LABORATORY, AHMEDABAD

- Studying quantum and statistical aspects of magnetism through the Curie-Langevin model.

21/05/2023 - 22/07/2023

SUMMER INTERNNISER, BBSR

Computed Thermal Composition of Hadron Yields and Partition Function $Z(T,V)$ in the Grand Canonical Ensemble using Python.

SUMMER TRAINEEINTER UNIVERSITY CENTRE FOR ASTRONOMY AND ASTROPHYSICS (IUCAA), PUNE

Conducted spectroscopic analysis of astronomical spectral graphs using QfitView Analysis.

STUDENTFAKIR MOHAN AUTONOMOUS COLLEGE, BALASORE

- Operated analog and digital circuits, e.g., amplifiers, oscillators, logic gates, flip-flops, and counters. Conducted circuit analysis and signal processing with function generators and multimeters. Built circuits and waveform analysis with oscilloscopes and CROs.
- Performed experiments on the Franck-Hertz experiment, and measurement of Planck's constant through the photo electric effect. Used spectrometers and other measurement tools to study atomic and quantum phenomena.

STUDENTFAKIR MOHAN (AUTONOMOUS) COLLEGE, BALASORE

- Conducted experiments involving laser optics, diffraction, and interference patterns. Gained hands-on experience with optical setups, alignment techniques, and measurement of optical properties of materials.

STUDENTFAKIR MOHAN (AUTONOMOUS) COLLEGE, BALASORE

- Studied Zeeman splitting in atomic spectra using a strong magnetic field. Analyzed the effects of the magnetic field on energy levels and the splitting of spectral lines in atomic transitions.
- Studied the properties of semiconductors and conducted experiments on the temperature dependence of conductivity, Hall effect, and ferromagnetism. Investigated Zeeman splitting in atomic transitions under the influence of a magnetic field.
- Performed data analysis using Python to plot and interpret experimental results.

EDUCATION & TRAINING

26/12/2025 - Current - WARANGAL, INDIA

PH.D (PHYSICS : QUANTUM INFORMATION PROCESSING & QUANTUM PHOTONICS)- NATIONAL INSTITUTE OF TECHNOLOGY, WARANGAL

29/11/2022 - 03/08/2024 - BALASORE, INDIA

M.SC. (PHYSICS)- FAKIR MOHAN (AUTONOMOUS) COLLEGE, BALASORE

Field(s) of study: Physics | **Final grade:** A | **Thesis:** High Energy Phenomenology Electron-Electron Anti Neutrino Scattering in Fermi Theory

05/06/2018 - 22/09/2021 - BHUBANESWAR, INDIA

B.SC (HONORS)- SRI JAYADEV COLLEGE OF EDUCATION AND TECHNOLOGY, NAHARKANTA

Field(s) of study:Physics | **Final grade:** A | **Thesis:** Theoretical Exploration of Raman Spectroscopy

08/08/2022 - 18/09/2022 - BHUBANESWAR, INDIA

WEB DESIGNING - DIGITAL DETA

Final grade: A

07/05/2021 - 28/05/2021 - DEHRADUN, INDIA

SAR DATA PROCESSING- INDIAN INSTITUTE OF REMOTE SENSING (IIRS)

22/08/2023

QUANTUM COMPUTING- MINDLUSTER

31/07/2023

INTRODUCTION TO MATLAB PROGRAMMING (FALL 2011)- MIT OPENCOURSEWARE

06/09/2023 - 25/09/2023

QUANTUM COMPUTING FOR NATURAL SCIENCES (WITH IBM QUANTUM)- HASSO PLATTNER INSTITUTE

- Simulating molecular dynamics on quantum computers
- Challenges and prospects of quantum simulation
- Common error sources and how to mitigate them
- Mapping the quantum description of molecules to the qubits of a quantum computer
- Computing ground state and excited state energies of molecules with quantum algorithms

Field(s) of study:Quantum Computing

MATLAB ONRAMP- MATLAB ACADEMY

MACHINE LEARNING ONRAMP- MATLAB ACADEMY

QUANTUM TECHNOLOGY SPIN QUBIT SEMICONDUCTOR MODELING AND SIMULATION- IMPULSE TECHNOLOGY

SIGNAL PROCESSING ONRAMP- MATHWORKS

NUMERICAL AND SCIENTIFIC PROGRAMMING- MINDLUSTER

MACHINE LEARNING WITH PYTHON- UDEMY

AI & COGNITIVE SCIENCE - UDEMY

QUANTUM COMPUTING (QOHORET-3)- QUCODE BITS TO QUBITS

● **SKILLS**

MATLAB | C | C++ | Wolfram Language | Object-Oriented Programming | Basic Data Analysis in Kaggle | Signal Processing | Image Processing | Numerical & Scientific Programming | AI & Cognitive Science | Supervised Machine Learning (Regression, Classification) | Python (Scipy, Numpy, Matplotlib, Pandas, Scikit-learn) | Python (Tensorflow and Keras) | Unsupervised Learning: Clustering (K-Means, Hierarchical), Dimensionality Reduction (PCA) | Neural Network | Convolutional Neural Networks (CNN) | Model Training using Deep Learning | Familiarity with Deep Learning models such as ANN, CNN, and RNN | XGBoost, Gradient boost, ensemble methods, ANN, CNN, RNN, LSTM, MLP | ANN, CNN, RNN, Keras, Pandas, Numpy, Scikit-learn | Data Handling | Quantum Algorithms | Quantum Gates & Circuits | QNN | Basic QML

Experimental Skills

Electronics (Analog & Digital Circuits, Network Analysis, Diode, Transistor, Opamp,etc..) | Optics (Interference, Diffraction, Laser-Based Experiments) | Measurement Techniques & Instrumentation | Solid State Physics (Zeeman Effect, Hall Effect Experiments, Four Probe Method, etc...)

Web Technology

HTML5 | CSS | WordPress Development | Bootstrap

Tools & Software

LANGUAGE SKILLS

Mother tongue(s): ODIA

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
HINDI	C2	C1	C2	C2	C1
SANSKRIT	A2	B1	A1	A1	A2
ENGLISH	B2	B2	B2	B2	B2

PROJECTS AND INTERNSHIP

15/05/2023 - 03/07/2023
Summer Research Intern at Physical Research Laboratory, Ahmedabad

- Completed summer internship at Physical Research Laboratory (PRL) on the Curie-Langevin theory of magnetism, exploring its quantum and statistical properties under Dr. Navinder Singh's guidance

22/05/2023 - 21/07/2023
Summer Research Intern at National Institute of Science Education and Research, Bhubaneswar

- I finished my internship under the guidance of Dr. Victor Roy at NISER, I delved into the intriguing intersection of "Hadron Resonance Gas and Lattice QCD" bridging theory and computation.
- Computed Thermal Composition of Hadron Yields and Partition Function $Z(T,V)$ in the Grand Canonical Ensemble using Python.

CONFERENCES & SEMINARS

15/02/2023 - 16/02/2023 Fakir Mohan (Autonomous) College, Balasore
National Conference on Application of Material Science in Emerging Fields - 2023

08/05/2023 - 15/06/2023
International Conference on Quantum information and Quantum technology 2023 Sponsored by the Indian Institute of Science Education and Research Kolkata & I-Hub Quantum Technology Foundation, Pune.

Physical Research Laboratory, Ahmedabad
Space Weather Workshop and Conference (ISWC-2)

SORA-VISTA 2025 (Society of Research and Applications (SORA))

HONOURS AND AWARDS

16/02/2023 Fakir Mohan (Autonomous) College, Balasore
I achieved the prestigious First Prize for my poster presentation on Density Functional Theory (DFT). At The National Conference on Application of Material Science in Emerging Fields - 2023

29/09/2023 Fakir Mohan (Autonomous) College, Balasore
Quiz Competition-2023

HOBBIES AND INTERESTS

Hiking & Outdoor Exploration, Fitness & Physical Training, Cycling, Sports (Cricket, Badminton, etc.).